

Problem Statement: AI-Driven Employee Attrition Prediction

Background

Employee attrition is one of the major challenges faced by modern organizations. High attrition rates can lead to:

- Increased recruitment and training costs
- Loss of experienced employees
- Reduced productivity
- Project delays
- Lower employee morale

A leading IT services company has observed an increase in employee resignations over the last two years. The Human Resources (HR) department currently relies on manual analysis and assumptions to identify employees who may leave the organization, which is inefficient and often inaccurate.

The management wants to adopt a data-driven approach using Artificial Intelligence and Machine Learning to proactively identify employees who are at risk of leaving the company.

Business Objective

Your task is to develop a supervised machine learning solution that predicts whether an employee is likely to leave the organization based on employee-related factors such as:

- Salary
- Overtime
- Job satisfaction
- Work-life balance
- Promotions
- Performance
- Department
- Experience
- Absenteeism
- Tenure

The solution should help HR teams:

- Identify high-risk employees early
- Understand the major reasons for attrition
- Improve employee retention strategies
- Reduce operational and hiring costs

Dataset

Dataset to be used:

[Employee Attrition Prediction Dataset](#)

Problem Type

This is a **Supervised Machine Learning – Classification Problem**.

Target Variable

Attrition

- Yes → Employee left the organization
- No → Employee stayed in the organization

Tasks to Perform

1. Business Understanding

- Understand the attrition problem
- Explain why employee retention is important
- Identify possible business impact of attrition

2. Data Understanding

Perform:

- Dataset exploration
- Missing value analysis
- Datatype analysis

- Duplicate checking
 - Statistical summary
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3. Exploratory Data Analysis (EDA)

Analyze and visualize:

- Attrition distribution
- Attrition by department
- Attrition by overtime
- Attrition by salary
- Attrition by job satisfaction
- Correlation between features

Students should provide meaningful business insights from the visualizations.

4. Data Preprocessing

Perform:

- Handling missing values
 - Encoding categorical variables
 - Feature scaling (if necessary)
 - Train-test splitting
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5. Model Building

Build and compare the following supervised learning models:

Mandatory Models

1. Logistic Regression
 2. Decision Tree
 3. Random Forest
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6. Model Evaluation

Evaluate the models using appropriate metrics such as:

- Accuracy
- Precision
- Recall
- F1-Score
- Confusion Matrix

Students should explain:

- Which model performs best
- Why certain metrics are important in business scenarios

7. Business Insights and Recommendations

Students must identify:

- Key factors influencing attrition
- High-risk employee profiles
- Departments with higher attrition risk
- Possible HR intervention strategies

Provide actionable recommendations for management.

ICT ACADEMY
K E R A L A